CHALLENGE

01

Seconds Ahead

Every problem on the road gives a warning first. Build the thing that catches it.

POWERED BY

PCB Cupid

Hardware kit partner · build it with the PCB Cupid kit

Hacktivate 2026 · 24-hour Hardware + AI Hackathon · Theme: Predictive Systems
14 Oct 2026, 11:00 AM to 15 Oct 2026, 11:00 AM · Pravrutti 2.0 (National Tech Fest)

Goals and Motivation

Meet Imran (one example of many)

Imran is 38 and drives an autorickshaw out of Peenya. Every morning at 7:40 he picks up two schoolchildren from a narrow cross road and turns onto the main road. A lorry is almost always parked at the corner. From the driver's seat, he can see its tyres and nothing else, so he does what every driver in India does: honks, noses out, and hopes.

One Tuesday a scooter is already there. The rider brakes, skids and goes down a metre from the auto's front wheel. Nobody is badly hurt, but the children are crying and Imran is shaking, because he knows it wasn't care that saved them. It was luck. And none of it was unpredictable: the scooter had been coming, fast and loud, for five seconds before he moved.

Imran's corner is just one example, not the challenge. Roads go wrong in a thousand everyday ways. A rider nodding off on the last stretch home. A scooter dropping into a pothole after dark. A car rear-ending a bike that braked suddenly. A junction that jams at the same time every evening. A parking lot that's full by the time you get there. A dip in the road that fills with rainwater. An engine that overheats in traffic. A tempers-rising moment at a signal. A breakdown on a flyover. An ambulance stuck behind a jam. Each is a different problem. What they share is that the warning signs were there seconds, minutes or days before, and nobody was told in time.

This is not a problem of missing technology. Speed, distance, sound, light, motion, temperature and patterns over time are all measurable with parts that cost a few rupees, and a small board can learn them. What is missing is **cheap products built for the roads ordinary people use every day that see the problem coming and warn someone, or act, before it happens.**

The Scale

India recorded **4,80,583 road crashes and 1,72,890 deaths in 2023**, about 20 deaths every hour (MoRTH, Road Accidents in India 2023). Use these numbers as evidence of how big the problem space is, and as a starting point for your own research, not as a list to pick from:

What the data shows (India, 2023)	Deaths	Source (MoRTH RAI 2023)
Two-wheeler riders killed (44.8% of all deaths)	77,539	Table 4.4
Car, taxi, van and LMV occupants killed	21,496	Table 4.4
Truck and lorry occupants killed	10,066	Table 4.4
Auto-rickshaw occupants killed	6,308	Table 4.4
Pedestrians killed (20.4%)	35,221	Table 4.4
Hit from behind, the biggest single collision type	36,804	Table 1.5
Crashes at junctions (19.9%); T-junctions alone 12,311	34,354	Table 3.6
Crashes on curved roads	22,263	Table 3.5
Over-speeding (68.1% of deaths)	1,17,682	Table 3.1
Riders and pillions killed without helmets	54,568	Table 3.3
Drivers and passengers killed without seat belts	16,025	Table 3.3
Use of a mobile phone while driving	2,884	Table 3.1
Crashes on pothole roads	2,161	Table 3.5

The deadliest window is **6 PM to 9 PM, with 20.8% of all crashes** (Table 7.3). Worldwide, road crashes kill about **1.19 million people a year** and are the leading killer of children and young people aged 5 to 29 (WHO Global Status Report on Road Safety 2023). And deaths are only part of it: jams, potholes, breakdowns, waterlogging and parking hunts cost crores of people time, money and temper every single day.

You are on these roads too. Somewhere on your own commute there is a moment like Imran's corner. This challenge asks you to find it, and to put yourself, and everyone like you, seconds ahead of it.

Three Failures. One Challenge.

Everyday roads fail ordinary people in three ways at once:

WARNINGS ARRIVE TOO LATE	Horns, brake lights, mirrors, signboards and speed breakers all work in the last second, or after it. By the time a rider sees the brake light, a driver spots the pothole or the jam shows up on a map, the moment to act has already passed. Almost nothing on the road says what is about to happen.
SMART ROAD TECH IS BUILT FOR THE FEW	Collision warnings, driver monitoring and smart traffic systems mostly ship in expensive cars and a handful of big-city junctions. The scooter, the auto, the old truck, the gali and the small-town signal get nothing.
THE ROAD IS DIFFERENT EVERY DAY	Indian roads are mixed and change by the hour: school rush, delivery peaks, cattle, wrong-side riders, rain, darkness, a new pothole, a tired driver at the end of a shift. Static signs and fixed rules can't know this corner, this hour or this rider. Nothing learns the road and the people on it.

Your Challenge

Pick any problem that happens on the road, and build a product with the PCB Cupid kit that predicts it before it happens.

Safety, traffic, potholes, crossings, parking, signals, crashes, breakdowns, waterlogging, road rage, emergency response, or something nobody has thought of — you decide. Your product can live in a helmet, on a handlebar, on a dashboard or steering wheel, inside a car, auto, bus or truck cab, at a junction or roadside, in a parking lot, on a wearable, or with the community that uses a road. **Your team lead is your primary user**, so pick a road problem your team really faces, or can genuinely experience, and test the product on yourselves. **You can choose to give your product a voice with ElevenLabs.** Voice is optional for this challenge, but if you use ElevenLabs voice your team is eligible for the Best Voice award.

A forecast, not an alarm

Your product must see the problem coming: how soon, how likely, or how a pattern is drifting towards trouble. "Something is close" or "you crashed" is a reaction. "That vehicle won't stop in time", "you're drifting towards a nod-off in about 5 minutes", "this junction jams in 15 minutes" or "this dip floods if the rain keeps up for 20 minutes" is a prediction.

Prediction is only half the job. The forecast has to reach the right person, in a form they can use in the moment, early enough to change what happens.

The Rules

- **Any road problem.** You choose the problem and the product, as long as it is about something that happens on the road.
- **Build with the PCB Cupid kit.** The core of your product — sensing, prediction and the warning or action — uses parts from the PCB Cupid kit (listed under Your Kit: PCB Cupid Components). Basic craft and mounting materials (cardboard, tape, foam, string, or a helmet, bag, handlebar or dashboard to mount it on) are fine.

- **The board does the thinking.** Your prediction runs on the kit's ESP32. A phone or laptop may play ElevenLabs audio triggered by the board (the kit has no speaker), show your logs or forecast plot, run a simulation that feeds your device during testing, and flash code. No camera, and no phone as the sensor.
- **A real prediction.** Time-to-event, an early-rising risk score or a learned pattern, as defined in The Prediction Requirement below. Crash detection after the fact and simple threshold alarms don't count on their own.
- **Be the user.** The team lead is the primary user, observes their own road experience, and tests the product on themselves (see Be the User below).
- **Test safely, always.** Self-testing happens on a stationary or parked vehicle, in a quiet campus lane while walking, on a table-top road model, or on a table-top rig or mock setup of your own. Never test in live traffic in a way that puts anyone at risk.

Building Blocks

Most predictive road products are made of these four pieces. They describe the thinking; the parts come from the PCB Cupid kit.

01 Be the User

Before you build: watch your own road. The team lead (and anyone else in the team who shares the experience) observes their own daily journeys — riding, driving, being driven, parking, waiting at a junction. Log every near-miss, scare or annoyance: a photo of the spot, a quick note of what happened and when, or a short voice memo recorded after you've stopped safely. Two or three honest entries beat a long imagined list.

During the 24 hours: test it on yourself. Wear it, fit it or mount it, and try it in safe ways: on a parked scooter or car, on a table-top road model, or on a table-top rig or mock setup. Every time, note what you tried, what the device predicted, what actually happened and what you changed.

At the end: be honest. Would you actually use this every day? What would make you switch it off? What would make you buy it?

Required: Bring a short **commute log** (your real near-misses or annoyances, with at least one photo, note or voice memo) and a mini **test log** of your own self-tests (see the example below), plus one sentence answering "Would I actually use this daily?"

What we tried	What it predicted	What actually happened	What we changed
Lead used the prototype for 20 minutes in our safe test setup	"Risk rising, event in about 4 min" at minute 14	The event came at minute 17; the warning was useful but too harsh	Made the warning calmer and shorter
Repeated the same test at different speeds	Warned on every run, even slow ones	Annoying; lead said they would switch it off	Added a rate-of-change check; slow runs now stay silent

Example of the log format only; the content is invented. Yours will look different.

02 Seeing It Coming

Find the signals that change before the problem arrives: a closing speed, an engine getting louder, headlights in the mirror, a head that starts to nod, a grip that loosens, a cabin getting hotter, a junction filling up, water rising, a time of day, a rider's habits, a spot's history. Capture them with the kit's sensors — ultrasonic, IR, microphones, LDRs, hall sensors with magnets, rotary encoders, the touch slider, the joystick, DHT11, the MPU6050 — combining two or more where it helps, and log real data early, because your forecast is only as good as what it has seen.

03 The Forecast

Turn the signals into a statement about the future: "impact risk rising, about 3 seconds", "you usually start drifting 40 minutes into a night ride", "this junction jams at 6:10 most days", "engine heat will cross the safe point in about 8 minutes at this rate". Show how far ahead and how sure you are. Simple, honest models often beat impressive-sounding ones.

Required: A real prediction, as defined in The Prediction Requirement below.

04 Acting in Time

Get the forecast to the person who can act on it, in a way that fits their moment: a buzz near a rider's ear, a glow at the edge of a driver's vision, a countdown on a 7-segment display, an LED strip that warns the vehicle behind, a 16x2 LCD at a parking gate, or an ElevenLabs voice played on a phone or laptop your board triggers. If it annoyed you during self-testing, it will annoy everyone. A warning you learn to ignore, or one that cries wolf, is worse than none.

Stretch Goals

- **Learns the road.** Your product adapts to a specific route, junction, rider or time of day, and gets better with use.
- **Warns both sides.** The person in danger and the person causing it both find out in time.
- **Talks to other units.** Two kit boards share what's ahead over BLE or Thread (ESP32-H2/C6).
- **Cheap enough to matter.** A price that a student, a rider, an auto driver or a small fleet would actually pay.
- **Voice.** Speak the forecast with ElevenLabs (credits for every team) where it can be heard — see Where Voice Fits below.

Where Voice Fits

A busy street is a hard place to hear anything, so think about **who** hears the voice, **when** and **how much**. These are places where a spoken message can genuinely be heard. What you build, and whether it speaks at all, is up to you. Every team gets **ElevenLabs credits** if it does:

- **Inside a vehicle cabin** — a car, auto, bus or truck driver can hear a short spoken prompt.
- **Through a helmet earpiece** — a rider can hear one short word or phrase, not a conversation.
- **Before a trip** — at home, at a hostel or at a depot, when someone has time to listen before setting out.
- **After a trip** — once the vehicle has stopped and the person can listen properly.
- **Where the road is managed** — a traffic marshal's post, a dashboard or a control room.

Keep it safe: short, calm, non-distracting prompts at the right moment. Never ask a rider or driver to look at a screen or hold a conversation while moving. The kit has no speaker, so play ElevenLabs audio through a phone or laptop (or an earpiece connected to it) triggered by your kit board, with the prediction still running on the board. Any language is fine.

The Prediction Requirement

This is a Predictive Systems hackathon. A product that reacts after the problem has arrived is an alarm, not a forecast.

Your device must make a **real prediction**. That means it forecasts something that has not happened yet, in one of these forms:

- **Time-to-event** — "about 40 seconds until this happens", updated as the situation changes.
- **An early-rising risk score** — a likelihood that climbs as precursors build up, well before the event itself.
- **A learned pattern** — the device learns what normal looks like for this person, place or machine, and flags the drift away from it before it becomes a problem.

Threshold alarms do not count. "Beep when the distance drops below 50 cm" or "buzz when the temperature crosses 40°C" is a reaction, not a prediction. A real forecast says what is coming and how soon, before any threshold is crossed.

Examples of what a real prediction looks like for this challenge:

- A seconds-to-impact estimate that rises as something closes in, and stays calm when the same thing is close but slowing.
- A risk score that starts climbing minutes before a person's behaviour drifts into danger, compared against their own learned baseline.
- A forecast of when a road condition will cross a problem point ("in about 15 minutes"), updating as the situation changes.

Required: Show what your product predicts, how far ahead and how sure it is, and show it working live in your demo.

Be Your Own First User

This challenge is built for and by real road users: you. Your team lead is the primary user, and the product should solve something that really happens to them on the road. You are welcome to also talk to someone who shares that road (a friend who rides the same route, the auto driver who takes you to college, a traffic marshal at the junction near campus), but your own experience and self-testing come first.

Required: Bring your **commute log** (real near-misses or annoyances, with at least one photo, note or voice memo), your **test log** from safe self-tests during the 24 hours, **one design change** your own use caused, and an honest answer to **"Would I actually use this daily?"**

The Demo Requirement

Hacktivate is judged on what works in front of the judges. There are no slides. Your demo is the device, your KiCad design and the person you built it for.

- **Live, no slides.** You present the working hardware on the table. A phone or laptop may show your sensor readings, forecast plot or serial monitor, but not a pitch deck.
- **KiCad circuit.** Show the schematic (and PCB layout, if you got that far) for the circuit you actually built. It should match the hardware on the table.
- **Working hardware.** A simulation, video or Wokwi model does not count as the live demo.
- **Show it working.** Run a live demo of your product doing its job: sensing, making its prediction and acting on it.
- **Demo video (required).** Record a short video (about 2 to 3 minutes) during the hackathon showing your working product making its prediction and acting on it. Submit it with your final submission before the 24-hour deadline (15 Oct 2026, 11:00 AM). Submission details will be shared by the organisers. The video is in addition to the live demo on the table, not a replacement for it.
- **Show your logs.** Your commute log and test log are part of the demo; keep them on a phone, laptop or paper on the table.
- **Bring the road to the table, your way.** Use whatever setup shows your product working on your problem. You choose it.

Required: All of the above: a live demo of about five minutes on the table, followed by questions, plus your 2 to 3 minute demo video submitted before 15 Oct 2026, 11:00 AM.

Your Kit: PCB Cupid Components

Challenge 01 products are built with these components. This is the full PCB Cupid kit, with quantities for the whole event (about one ESP32 per team), so ask at the help desk.

Component	Quantity
ESP32 Microcontrollers H2	10
ESP32 Microcontrollers C6	10
LEDs (Red, Green, Blue - Individual)	120
RGB LEDs	100
Toggle Switches	10
Push Buttons (12×12 mm)	32
Jumper Wires (Male-Female)	10
Jumper Wires (Female-Female)	10
Jumper Wires (Male-Male)	10
Breadboards	20
IR Sensors	13
Rotary Encoders	17
LDR (Light Dependent Resistor)	30
Resistors	100 each × 5 values
Condenser Microphones	20
5V 1A Power Adapters	10
DC Power Adapter Breakout Boards	10
Ultrasonic Sensors	15
Hall Sensors	19
Magnets	35
MPU6050 - Gyro / Accelerometer	5
DHT11 (Temperature / Humidity)	10
Potentiometer 1K	15
16x2 Display	10
Touch Slider	10
Buzzer	10
Seven Segment Display (Common Anode)	10
Joystick	10
RFID	5

- **Ultrasonic sensors are fewer than teams**, so have a plan B.
- **ESP32-H2 has no Wi-Fi.** Use an **ESP32-C6** if your board needs to trigger ElevenLabs voice on a phone or laptop.
- **Not in the kit:** speaker, camera, motor or vibration motor, relay, GPS. Craft and mounting materials are fine.
- **Provided by organisers:** ElevenLabs credits for every team.

Resources

Tools and Methods

- **KiCad** for the schematic and PCB layout.
- **Arduino IDE / ESP-IDF / PlatformIO** for the ESP32-H2 and ESP32-C6.
- **Simple, explainable methods first** — time-to-collision from distance and speed, rate of change, moving averages, Kalman filters, regression over a sliding window, anomaly scores against a learned baseline.
- **Tiny ML on the ESP32** — TensorFlow Lite for Microcontrollers or Edge Impulse, if a small learned model helps.
- **Open data** — MoRTH's Road Accidents in India report and city crash data can help you find where and when your problem is worst.
- **ElevenLabs** credits for every team, if your product speaks (audio plays on a phone or laptop triggered by your board).

What Makes a Strong Submission

Strong Submissions...	Weak Submissions...
— Be the user. The lead lives with this road problem, and the team can show the near-misses or annoyances from their own log that started the idea.	— Build for an imagined user and invent their needs the night before.
— Test it on yourselves, safely. A test log shows what you tried, what the device predicted, what happened and what you changed.	— Never try it outside the code editor, or only test it once, right before judging.
— Iterate from your own use. At least one real design change came from wearing, mounting or using the product.	— Keep the first version unchanged because "it works on the bench".
— Predict before it happens. The forecast leads the problem, states its lead time and confidence, and works live in the demo.	— Detect a crash after it happens, or beep at a fixed threshold, and call it predictive.
— Make the kit do the work. Kit sensors feed a prediction that runs on the ESP32, and kit outputs (or a board-triggered voice) deliver the warning.	— Lean on a phone or laptop for the sensing or the prediction, and use the kit only for an LED.
— Be original and honest. Your own idea, plus a straight answer to "Would I actually use this daily?", including what still annoys you.	— Build the most obvious idea without asking whether it solves your problem, or claim it is perfect and everyone will love it.
— Demo it live, and on video. Working hardware on the table, plus a short demo video submitted on time.	— Show slides, a video or a simulation instead of a working product on the table.

Evaluation Criteria

Every challenge is judged out of 20 marks, with extra marks for creativity.

Criterion	Marks	Weight	What judges look for
KiCad design	5	25%	A clean schematic for the hardware you built, matching the device on the table; sensible power and signal design, built from PCB Cupid kit parts.
Hardware concept and working (incl. prediction quality)	7	35%	Is the road problem real, found in your own experience? Is the prediction real, early and honest about confidence? Is the product original and built with the kit, does it act on the forecast in time, and did it improve through your own testing?
User understanding	3	15%	Be the User evidence: a real commute log, a safe self-test log, design changes driven by your own use, and an honest "would I use this daily?" reflection.
Live demo	5	25%	Live, no slides; clear in about five minutes; the product works in front of the judges; demo video submitted on time.

Ground Rules

- **PCB Cupid kit plus craft and mounting materials** for the core build of Challenge 01. A phone or laptop may play board-triggered voice, show logs, run a test sim and flash code only.
- **Test safely.** Self-test on a stationary or parked vehicle, on a table-top model or driving rig, or walking in a quiet campus lane. Never test in live traffic in a way that puts anyone at risk, and never record voice memos or notes while riding or driving. Anything that acts on a real vehicle (brakes, throttle, steering) must be demonstrated on a model.
- **Assistive and early-warning aids only.** No medical diagnosis claims. "Your posture is drifting" is fine; "you have a spinal condition" is not.

Why This Matters

India loses more people on its roads than almost any country on Earth, and crores more lose time, money and calm to jams, potholes, breakdowns and near-misses every day. Most of them ride scooters, drive autos and trucks, or take whatever the road gives them, with nothing but eyes, ears and a horn. And every problem arrives as a surprise, even though the warning was there seconds, minutes or days before.

Sensors are cheap, boards can learn, and every one of you uses these roads. If you can build a product from a box of kit parts that sees the problem first and gets the warning to the right person in time, you won't just win a hackathon. You'll have built something that could keep someone you know safe.

Appendix — Contact & Additional Resources

Challenge contact: Ganesh S, Chair, IEEE SB, Ramaiah University of Applied Sciences · voxmastery@ambugo.tech · +91 79753 24769

Web: pravrutti.tech · msruas.ac.in

- Kit help desk: on site at Pravrutti 2.0 for PCB Cupid kit parts and replacements.
- Road safety data: Ministry of Road Transport and Highways (MoRTH), Road Accidents in India 2023 — Tables 1.5, 3.1, 3.3, 3.5, 3.6, 4.4 and 7.3. morth.gov.in
- Global context: WHO Global Status Report on Road Safety 2023 (who.int).
- KiCad documentation: docs.kicad.org · ESP32-H2 and ESP32-C6: docs.espressif.com · ElevenLabs: elevenlabs.io/docs